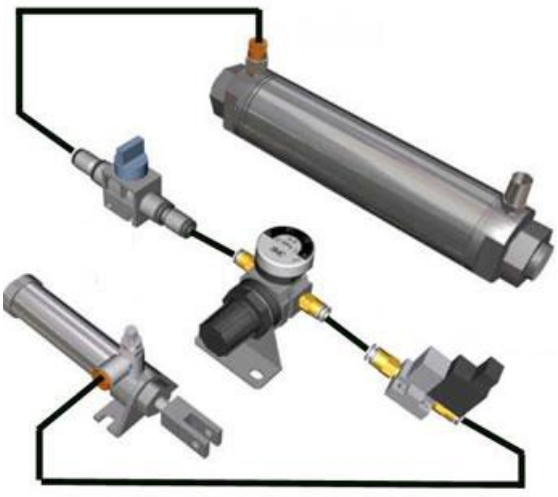
ENGINEERING SCIENCE

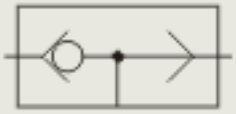
# Mechanisms and Structures: Pneumaics



**Things you should already know from National 5 Engineering Science.**

You should also know that we use the term ACTUATOR, not activator.

## Pneumatic Symbols

| | | | |
|---|---|---|---|
|  | Main Air Supply |  | Roller activator |
|  | Exhaust Air |  | Diaphram activator |
|  | Pilot Air |  | Solenoid activator |
|  | Air pipes Connected |  | Spring return |
|  | Air pipes not connected |  | Single Acting Cylinder |
|  | Air Bleed |  | Double Acting Cylinder |
|  | 3/2 Valve |  | 5/2 valve |
|  | Push Button activator |  | Reservoir |
|  | Plunger activator |  | Unidirectional Restrictor |
|  | Lever activator |  | Shuttle Valve |

# Introduction

Pneumatics use compressed air to do work. Pneumatic systems are common and are used in industry, to move objects off a conveyor or hold/clamp parts of an assembly together.







Footballs and bike tyres are pumped up with compressed air, and the air stays inside or slowly leaks out, like a balloon with a small hole. Most pneumatic systems rely on a constant supply of compressed air from a compressor.

# Uses:

The National 5 course explained the uses, advantages and disadvantages and different types of actuators in an air-controlled system. Higher ES investigates in depth operation of the systems.

# Health and Safety

Pneumatic devices and systems are safe, clean and reliable but we must learn to use them properly to avoid accidents. When using pneumatic equipment, we must always follow these safety rules:

1. Do not blow compressed air at anyone.
2. Securely connect all airlines before turning on mains air supply.
3. Always turn off the main air supply before changing a circuit.

# Cylinders

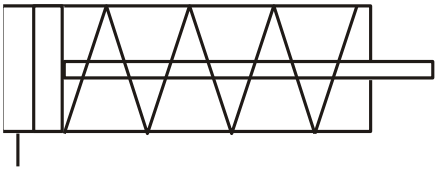
Pneumatic equipment can be split up into two basic categories of cylinders and valves.

Cylinders are the 'muscles' of pneumatic systems as they are used to move, hold and lift objects. They can even be used to operate other pneumatic components. Cylinders are operated by compressed air and they covert the stored energy into linear motion.

There are two types of cylinder that we will be using: a single acting cylinder and a double acting cylinder.

## Single Acting Cylinder

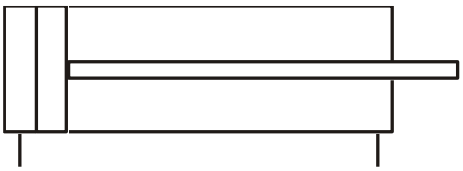
A single acting cylinder requires only one air supply. If we supply compressed air to a single acting cylinder, the air pushes against a piston inside the cylinder and causes it to ***outstroke***. When the piston has fully outstroked it is said to be ***positive***. If we stop the supply of air then a spring inside the cylinder causes the piston to ***instroke*** to its starting position and the piston is said to be ***negative***. The symbol for a single acting cylinder is shown below.



Single acting cylinders are used in situations where their simple design and small size are an advantage. They produce small outstroke forces and their use is limited to operations that only require use of the outstroke. This is because the spring produces only enough force to instroke the piston.

## Double Acting Cylinder

A double acting cylinder requires two air supplies, one to outstroke the piston and the other to instroke the piston. Compressed air is applied to one side of the piston then other to make it move positive and then negative. The symbol for a double acting cylinder is shown below.



Double acting cylinders are used more often in pneumatic systems than single acting cylinders. They are able to produce greater forces and we can take advantage of both the outstroke and instroke forces to lift and move objects.

*Cylinders will be described as '+' or '–'. The '+' means it is outstroking and the '–' means the cylinder rod is instroking. For example: A+, B-*

# Valves

Valves provide the necessary control of the compressed air being supplied to the cylinder. They can turn the air on or off, change the direction that the air is flowing or even slow down the airflow. The most common type of valve is the 3/2 valve.

## 3/2 Valve

A 3/2 valve is so called because it has 3 ports or connections and two states of operation. The ports are always numbered in the same way.

**Port 1** – main air supply connection

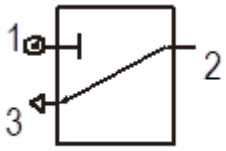
**Port 2** – output connection to other components

**Port 3** – exhaust

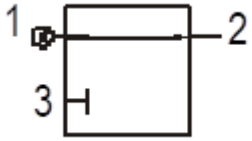
The valve has two states of operation. One state prevents air from being supplied to other components and the other allows the air to flow freely.

**State 1 – Off / Unactuated state**

In this state, the main airflow through the valve is blocked and so air is unable to reach other components such as cylinders. However, any air within the cylinder is able to exhaust through the valve and this will allow the cylinder to return to its original position. Study the symbol below and ensure that you understand how the air flows through the valve.




State 1

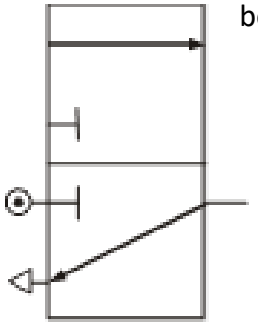



State 2

**State 2 – On / Actuated state**

In this state, the main air supply is able to flow freely through the valve and supply components such as cylinders with air. Study the symbol below and ensure that you understand how the air flows through the valve.

The complete symbol for a 3/2 valve combines both states and is usually drawn in the off or unactuated state. The complete symbol is shown below.



## 5/2 Valves

The problem with controlling a double acting cylinder with two 3/2 valves is that when neither valve is actuated, no air is being supplied to the cylinder to hold the piston in place. Any air within the cylinder is able to exhaust through port 3 on each valve. This means that any force or effort placed on the piston will make it move easily. A further disadvantage is that the 3/2 valve needs to be actuated until the double acting cylinder has fully outstroked or instroked. Releasing the valve before the stroke is complete will result in the piston stopping short of its final position.

We have greater control over a double acting cylinder if we control its outstroke and instroke using a 5/2 valve. This valve has 5 ports and two states of operation. The ports are always numbered in the same way.

**Port 1** – main air supply connection

**Port 2** – output connection to cylinder

**Port 3** – exhaust

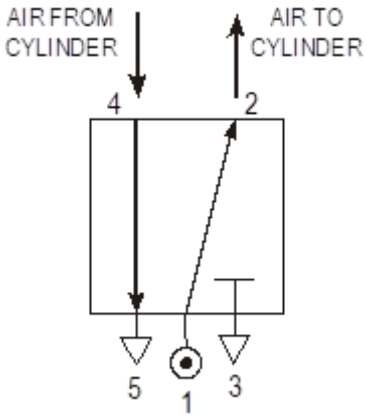
**Port 4** – output connection to cylinder

**Port 5** – exhaust

The valve has two states of operation. One state supplies air to outstroke a double acting cylinder and the other state will cause it to instroke.

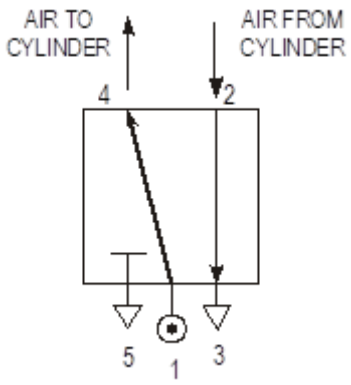
State 1 - Instroke

In this state, the main air flows through the valve from port 1 to port 2. Any air within the cylinder is able to exhaust through the valve from port 4 to port 5. In this state, a 5/2 valve will cause a double acting cylinder to instroke or hold the piston in the negative position as air is always being supplied to the cylinder. Study the symbol opposite and ensure that you understand how the air flows through the valve.

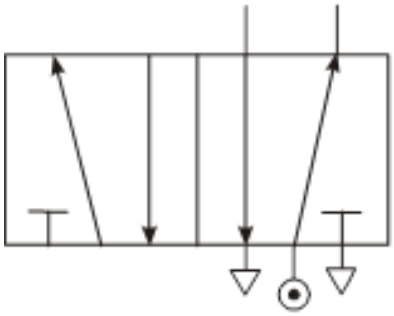



State 2 – Outstroke

In this state, the main air flows through the valve from port 1 to port 4. Any air on the other side of the piston is able to exhaust through the valve from port 2 to port 3. In this state, a 5/2 valve will cause a double acting cylinder to outstroke and hold it in the positive position. Study the symbol opposite and ensure that you understand how the air flows through the valve.

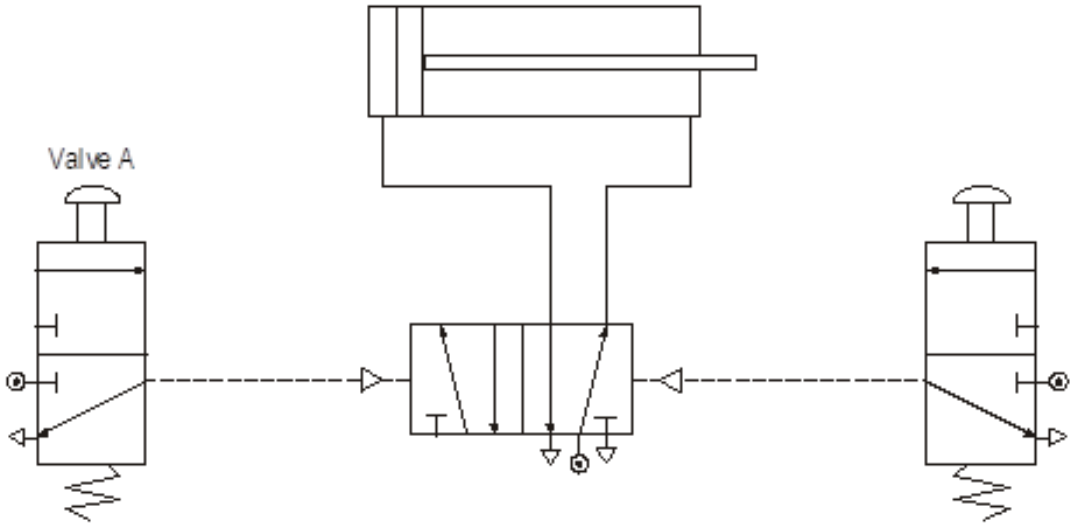


The complete symbol for a 5/2 valve is shown. The cylinder, main air and exhaust lines are only ever drawn on one half of the symbol depending on which part is active in the circuit. It is usual, however to draw the symbol in such a way as it would hold a double acting cylinder in the negative position.



## Pilot air

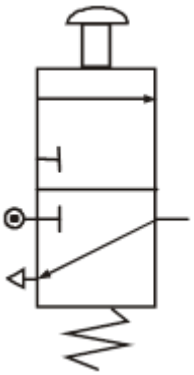
5/2 valves can be operated or actuated in the same way as 3/2 valves however, the most common way of actuating a 5/2 valve is pilot air. A pilot air controlled 5/2 valve will change 'state' when a shot of air signal acts at either end of the valve. This brief air signal (shot of air) is most often supplied from a 3/2 valve. In the example shown below, the button on Valve A only needs to be pressed momentarily in order to change the state of the 5/2 valve. The 5/2 valve will supply the double acting cylinder with air in order for it to outstroke.




Notice that the pilot air lines to the 5/2 valve are drawn as **broken or dashed lines** to distinguish them from the other air lines in the circuit.

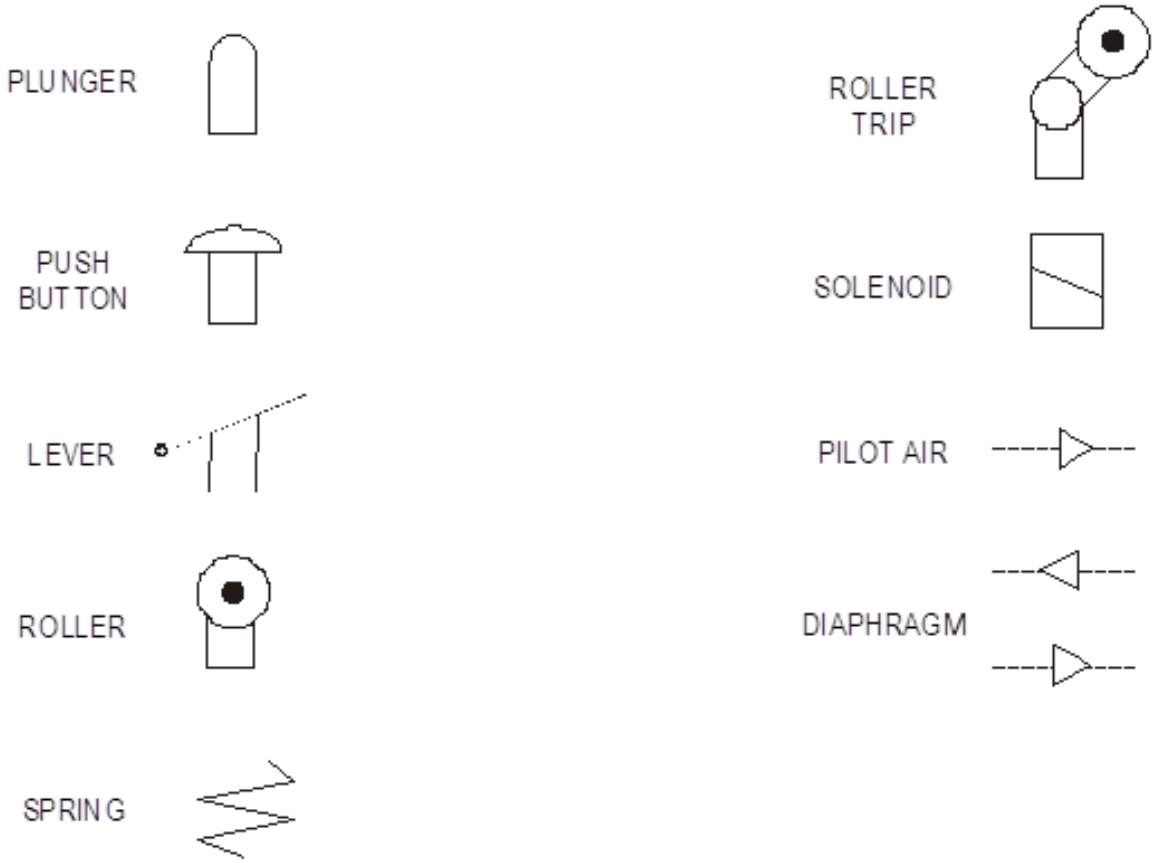
## Actuators

There are a number of different ways that we can change the state of the 3/2 valve. The most common way is by using a push button. A push button is useful for quick and easy starting of a sequence or as a safety circuit breaker. By pressing the button, the valve changes to the actuated state and allows main air to flow through to other components. If we release the button, a spring inside returns the valve to its off state.



The symbol for **a push button, spring return 3/2 valve** is shown.

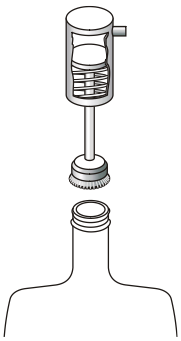
Below is a list of the most common types of actuators. They are always drawn onto the standard symbol for the 3/2 valve.

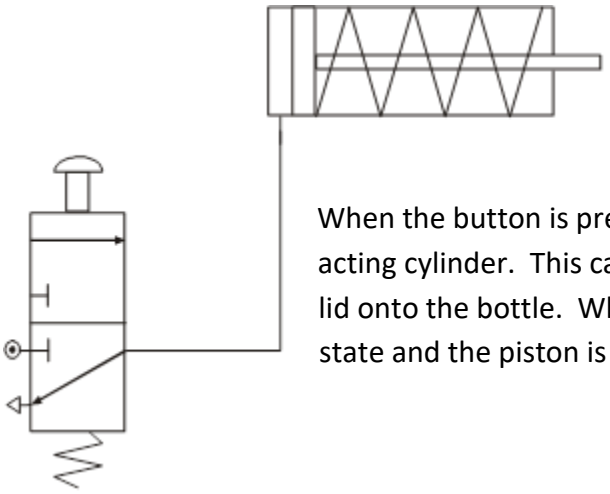



Not all of these actuators are intended to be used by a human operator. For example, an outstroking piston from a single or double acting cylinder actuates the roller trip. The solenoid uses an electrical pulse to change the state of the valve and this allows us to use combined electrical and pneumatic systems.

## Simple circuits

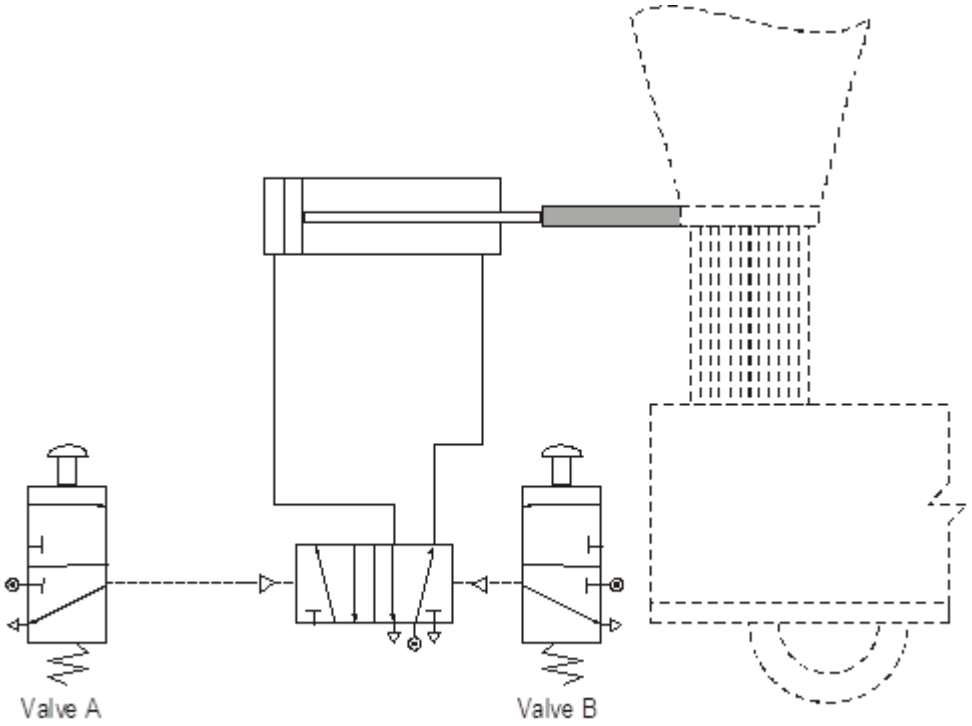
A bottling plant uses a single acting cylinder to press lids onto bottles. The cylinder is controlled by a push button, spring return 3/2 valve. An operator presses the button when the bottle is in the correct position. The pneumatic circuit is shown below.





When the button is pressed, the valve changes state and supplies air to the single acting cylinder. This causes the piston to outstroke with enough force to press the lid onto the bottle. When the button is released, the valve returns to its original state and the piston is able to instroke ready for the process to begin again.

**1.** The circuit below opens and closes the door of a grain hopper. In your jotters, describe how the circuit operates. When valve A is actuated…….



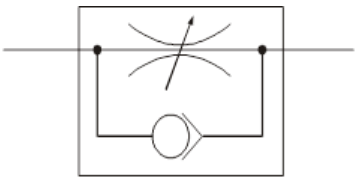

Air bleed

Diaphram

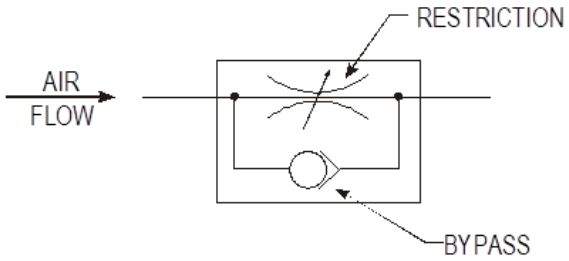
## Flow control valves

You will have noticed in all the circuits you have built so far, that the cylinders operate very quickly. Sometimes this is exactly what we want but very often it would be dangerous or impractical for the cylinder to work this fast. For example, think back to the exercise where a basket was lowered into a chemical solution. If this happened too fast then the chemical would splash out of the container. In this case, we would want the cylinder to raise and lower the basket slowly.

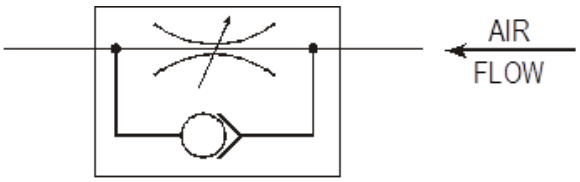
We can slow down the airflow to a cylinder by using flow control valves. The most common type is the unidirectional restrictor. This type of valve slows down the airflow in one direction only. The symbol is shown below.



When air flows into the restrictor in the direction shown below, some of the air takes the bypass route and forces the ball to seal the path. The air is then forced to pass through the restriction and the flow is slowed down. We can alter the flow by adjusting the small screw on the top of the restrictor.

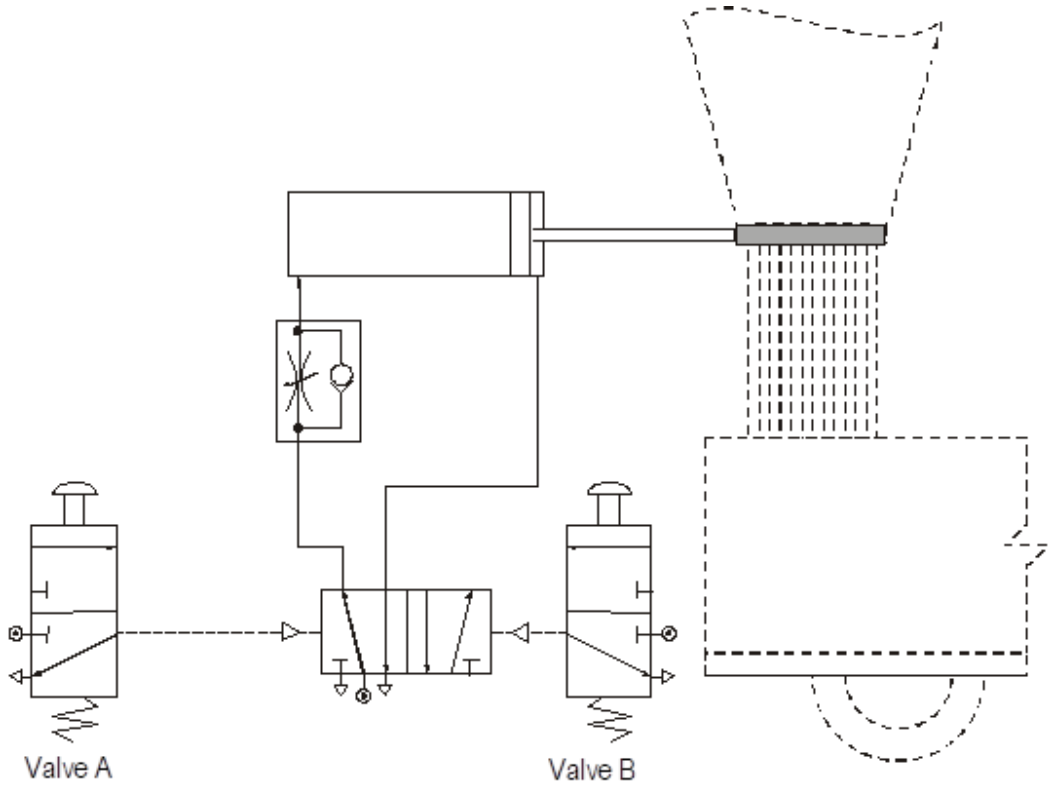



When air flows in the opposite direction into the restrictor, most of the air takes the bypass route and forces the ball to open up the path. This allows the air to flow freely and unrestricted.




If we want to restrict the stroke of a cylinder, it is normal practice to restrict the exhaust air coming from the cylinder rather than the main air being supplied to the cylinder. This provides a much smoother motion in the piston.

Remember our example with the hopper door. It would be much more practical for the door to open slowly, as this would allow the sand to fall more gently into the wagon below. We would still want the door to close quickly so that the process could begin again as soon as possible.

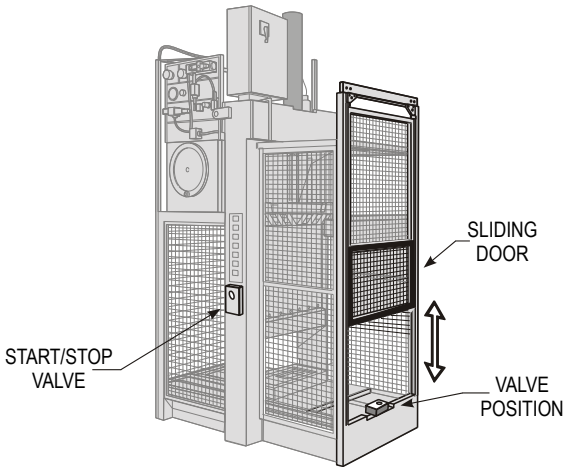



**2.** Study the new circuit diagram below.

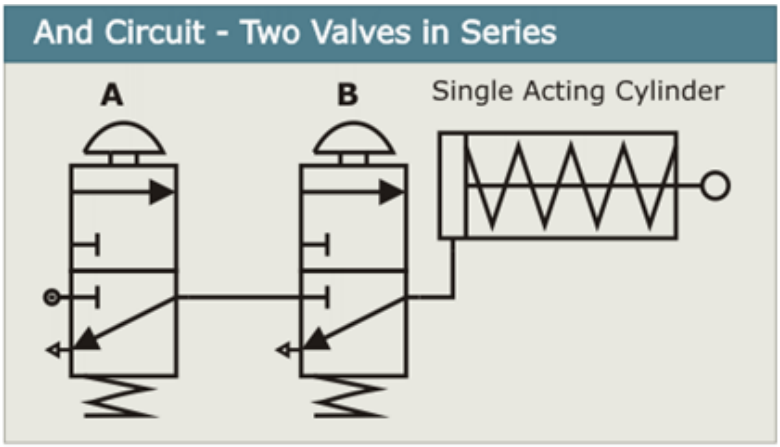
To open the door of the hopper slowly, we need to control the speed of the instroke. When Valve B is pressed, the pilot air 5/2 valve changes state and supplies the air to instroke the double acting cylinder. However, air is trapped behind the piston and this must escape or exhaust in order for the piston to move. The restrictor is connected in such a way that the exhausting air must pass through the restriction. This slows down the exhaust air and so slows down the speed that the cylinder can instroke.

## AND Control

We can create basic control circuits using pneumatic equipment relatively easily. On many production lines, operators become careless because they are so familiar with the machine that they constantly use. One way to avoid accidents happening is to fit the machine with a guard system, where the machine will only operate if the guard is in position and the machine is switched on by the operator.




This type of control can be achieved if we connect two 3/2 valves together in series.

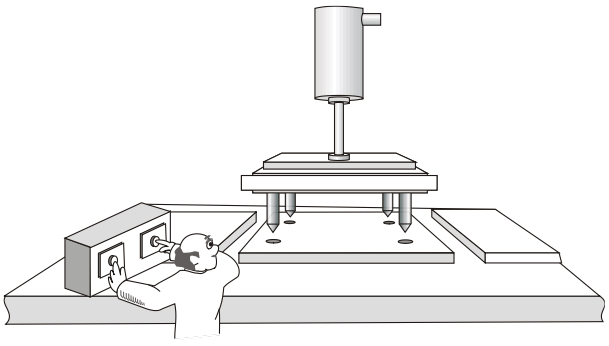



The single acting cylinder in this circuit will only outstroke if both 3/2 valves are actuated at the same time. This is because Valve B is supplied with air coming from Valve A. Pressing the button on Valve A supplies air to Valve B and if we press the button on Valve B at the same time, this allows air to flow into the single acting cylinder and cause it to outstroke. We can summarise the behaviour of this type of circuit in a truth table.

| VALVE A | VALVE B | CYLINDER |
|---|---|---|
| OFF | OFF | INSTROKE |
| ON | OFF | INSTROKE |
| OFF | ON | INSTROKE |
| ON | ON | OUTSTROKE |

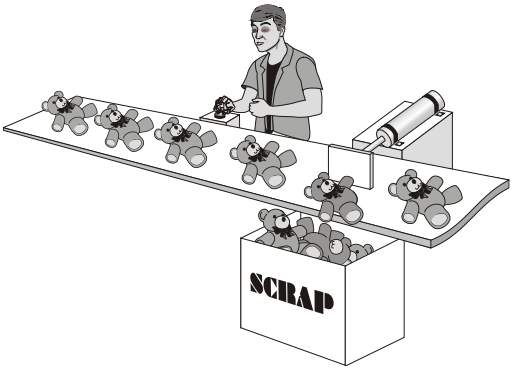
A machine in a factory is very difficult to guard without affecting how the machine carries out its task. The manager is very concerned that the operator may have an accident as a result. In order to prevent this, the manager wants to redesign the start/stop system of the machine so that the operator needs to press two buttons before the single acting cylinder outstrokes.

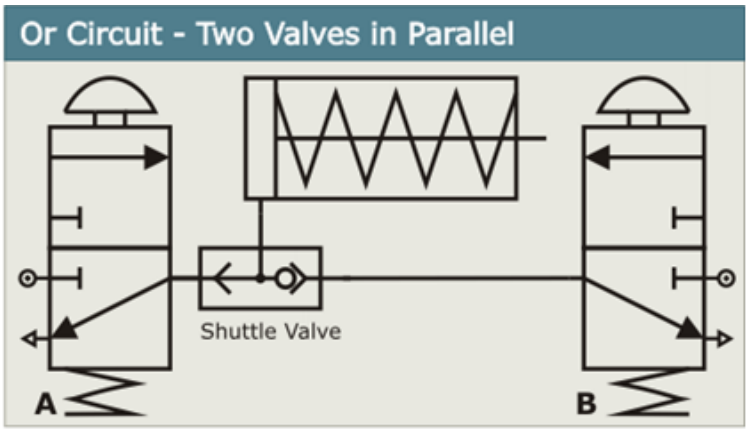
**3.** In your jotters, sketch a pneumatic circuit which would carry out this function.



## OR Control

In many industrial applications, pneumatic systems are controlled by either of two operators. They may be working together on the same process or controlling a process from two different positions. Quality control checks on production lines usually involve two or more people. Damaged or unsatisfactory goods are rejected if one or other person detects a fault and the goods are removed from the line by a single acting cylinder.

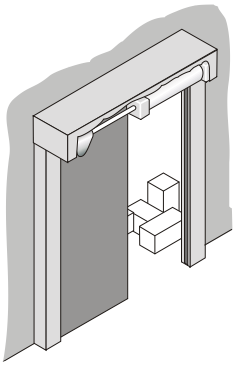


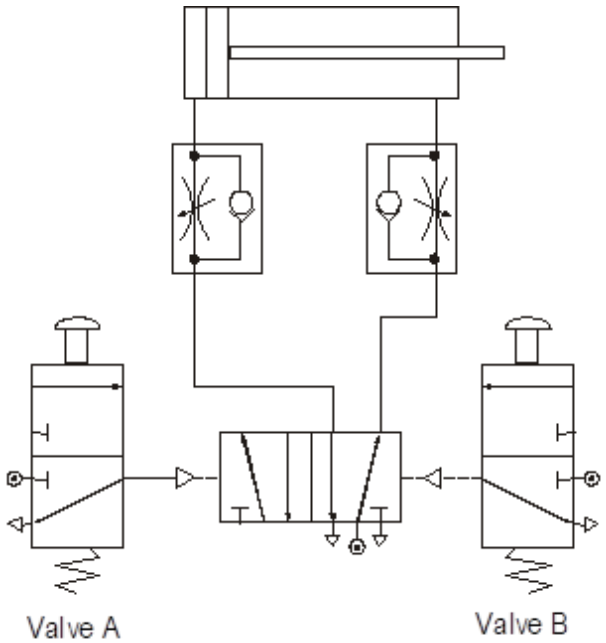




This type of control requires two 3/2 valves to be connected together in parallel and the use of an additional component called a shuttle valve. A shuttle valve is a very simple device but it is essential to how an OR control circuit works. If the button on Valve A is pressed, the ball in the shuttle valve is blown across towards Valve B and air is directed towards the single acting cylinder. This will outstroke the piston. If the button on Valve B is pressed, the ball is blown across towards Valve A and air is directed towards the cylinder. This also causes the piston to outstroke. We can summarise the behaviour of this circuit in a truth table.

| VALVE A | VALVE B | CYLINDER |
|---|---|---|
| OFF | OFF | INSTROKE |
| ON | OFF | OUTSTROKE |
| OFF | ON | OUTSTROKE |
| ON | ON | OUTSTROKE |

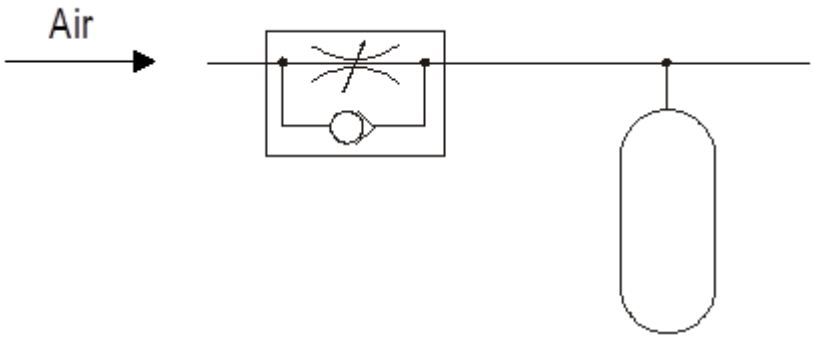
**4.** A warehouse door is controlled by the pneumatic circuit as shown opposite. In your jotter, describe the operation of this circuit.





## Time delay circuits

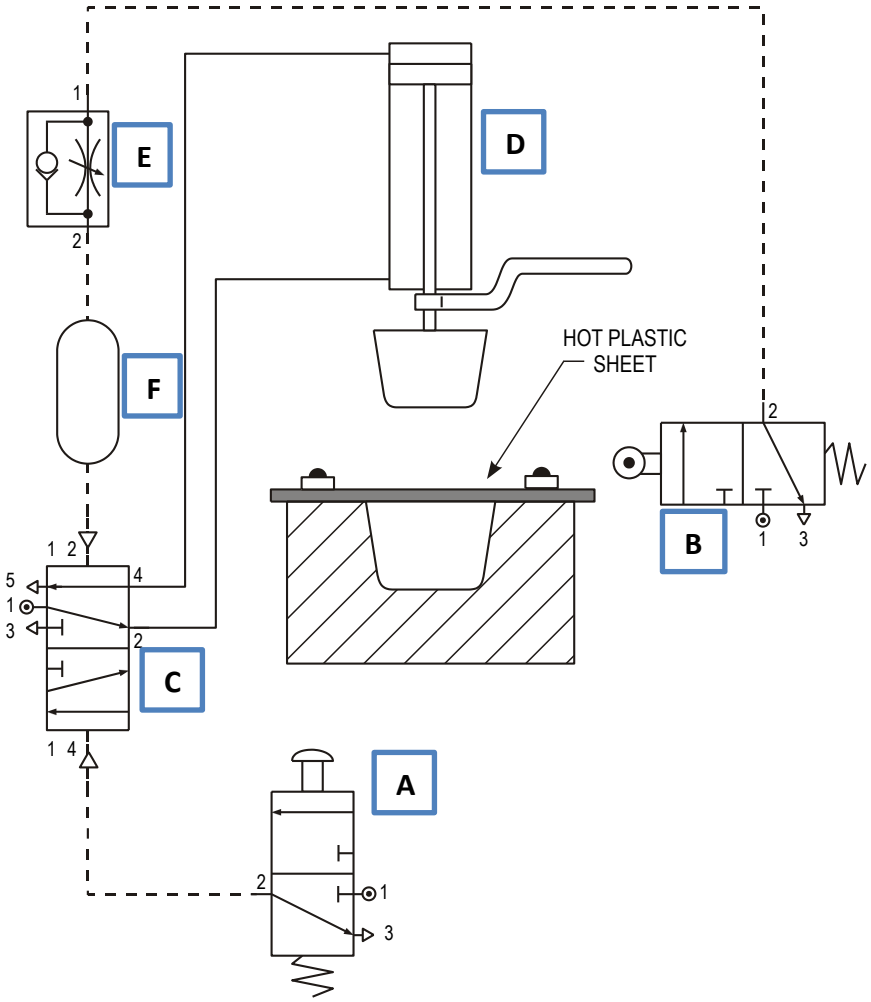
Sometimes we require a time delay or pause in a circuit before something else happens. Time delays are easily achieved with the use of two additional components inserted into the system at the point where the delay is required. The components required are a unidirectional restrictor (which we have already used) and a reservoir. The arrangement is shown below.



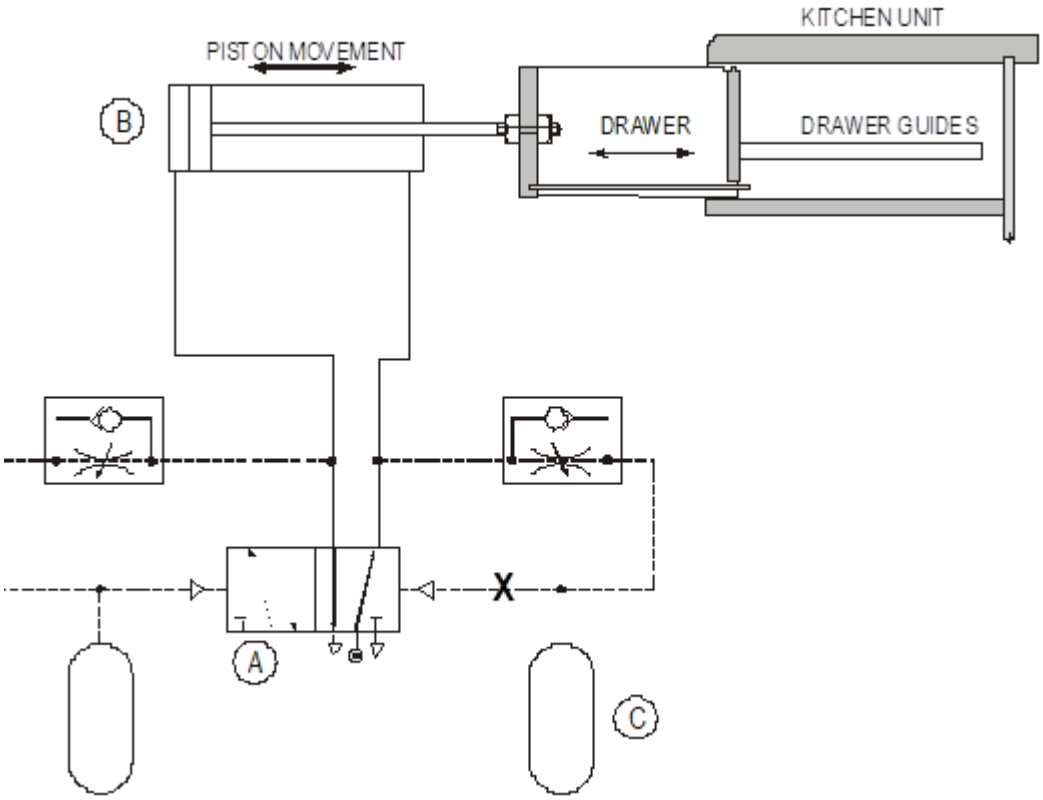

A reservoir is simply an empty container. When placed in a circuit it takes some time to fill up with air and restore the working pressure of the circuit. The time it takes to fill creates the pause or delay. We can change the length of the delay in two ways. The first is to use a reservoir of a different size as large reservoirs take longer to fill. The second way is to adjust the flow of air into the reservoir using a unidirectional restrictor. We can slow down the airflow using the restrictor so that the air takes a much longer time to fill the reservoir.

Time delay circuits can be useful in clamping or pressing applications where pressure is applied by a cylinder for a set time. Such a circuit can be used to make plastic containers. This type of pressing system makes use of a double acting cylinder to press a hot plastic sheet into a mould. The plastic needs to be held in position for several seconds to allow it to cool.

**5.** In your jotters, describe how the circuit operates.

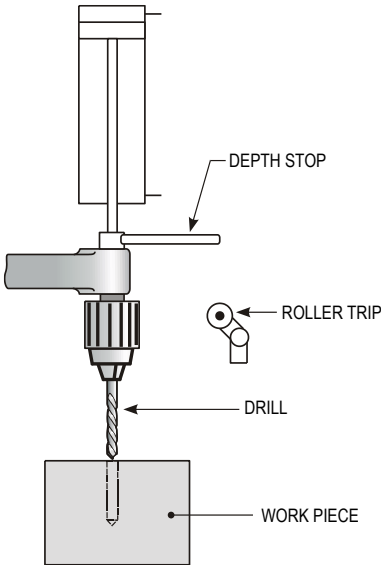


**6.** A pneumatic system for testing drawers is shown below. In your jotters, describe how the circuit operates



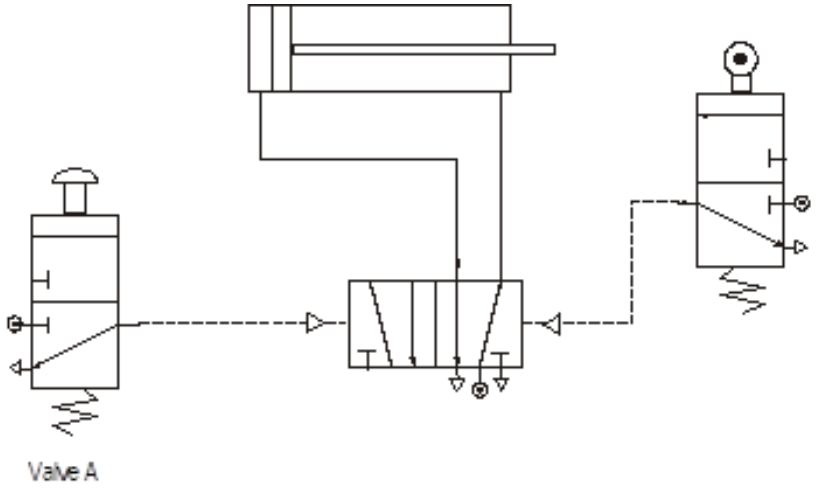

## Semi-automatic circuits

In a semi-automatic circuit, either the instroke or outstroke of the piston is automatic. The most commonly used semi-automatic circuits involve an operator starting the circuit and when the piston is fully outstroked, it actuates a valve that controls the instroke. We have come across examples of semi-automatic circuits in the course already.




On some production lines, drilling operations are started by an operator who is able to ensure that the component to be machined is in the correct position. When the hole has been drilled to the correct depth, a roller trip is activated which automatically instrokes the cylinder.

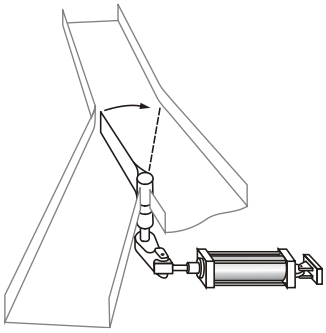
**7.** The pneumatic circuit that controls this operation is shown below. In your jotters, describe how the circuit operates.



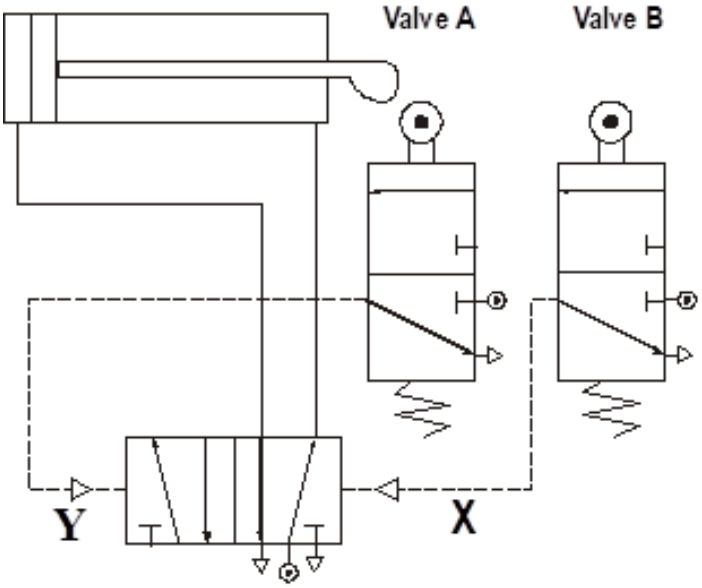
## Automatic circuits

Automatic circuits can perform tasks over and over again without the need of input from an operator. These circuits make use of actuators such as a roller trip and plunger to sense the position of the piston. We have come across an automatic circuit already in the course.

Automatic control circuits are used in applications that require reciprocating motion of the piston. For example, a production line requires the finished products to be directed towards two loading bays ready for shipping out to customers. A double acting cylinder controls the movement of a chute so that every second container is sent to each bay.



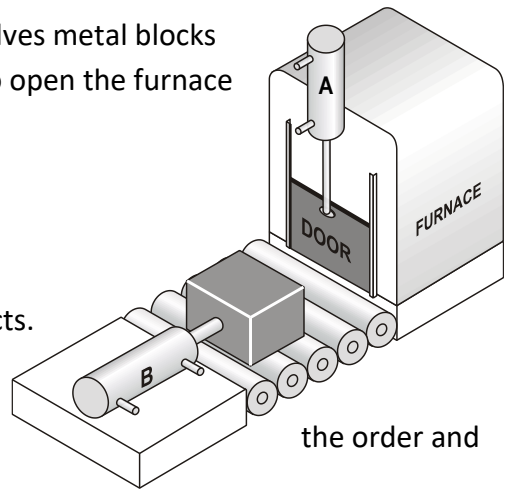
The pneumatic circuit that controls this operation is shown below. Notice that roller trip, spring return 3/2 valves are used to detect when the piston is positive and negative. In your jotters, describe how the circuit operates.




## Sequential circuits

Many pneumatic systems and machines are designed to perform a range of tasks in a certain order or sequence. This usually involves the use of two or more cylinders working together to complete the task.

For example, a company has automated its production line that involves metal blocks being placed in a furnace for heat treatment. One cylinder is used to open the furnace door and another pushes the metal blocks into the furnace.




The sequence of operations for this process is:

1. An operator pushes a button to start the process.
2. The furnace door is opened.
3. The block is pushed into the furnace and the piston retracts.
4. The furnace door is closed.
5. The sequence stops.

For this system to work successfully, we need to fully understand the order and movement of cylinders A and B.

**Stage 1:** Cylinder A instrokes to raise the furnace door.

**Stage 2:** Cylinder B outstrokes and pushes the metal block into the furnace.

**Stage 3:** Cylinder B instrokes.

**Stage 4:** Cylinder A outstrokes and closes the furnace door.

It can be summarised by the following: **Start**, **A-**, **B+**, **B-**, **A+**, **Stop**

The pneumatic circuit that carries out this operation is shown below.

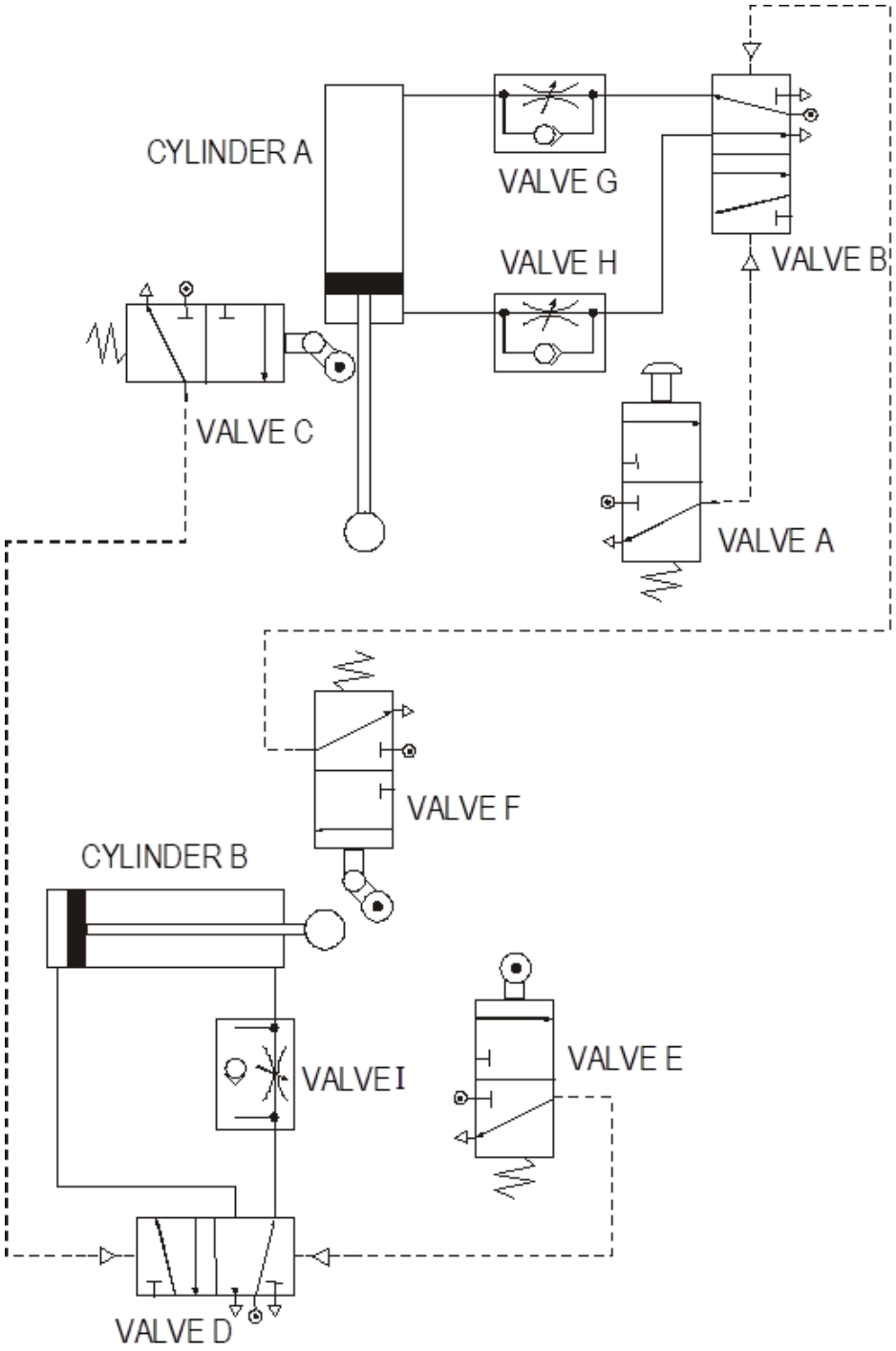


Diagram fully explained

- The system begins by actuating Valve A.
- This changes the state of Valve B and causes Cylinder A to instroke, raising the door.
- When fully instroked or ***negative***, the piston trips Valve C and this sends a signal to Valve D.
- This 5/2 valve changes state and sends Cylinder B ***positive***.
- When fully outstroked, the piston trips Valve E and the cylinder instrokes.
- When ***negative***, Valve F is actuated and causes Cylinder A to outstroke and stay in the ***positive*** position.
- The system stops and waits for a signal from Valve A.

## 8.

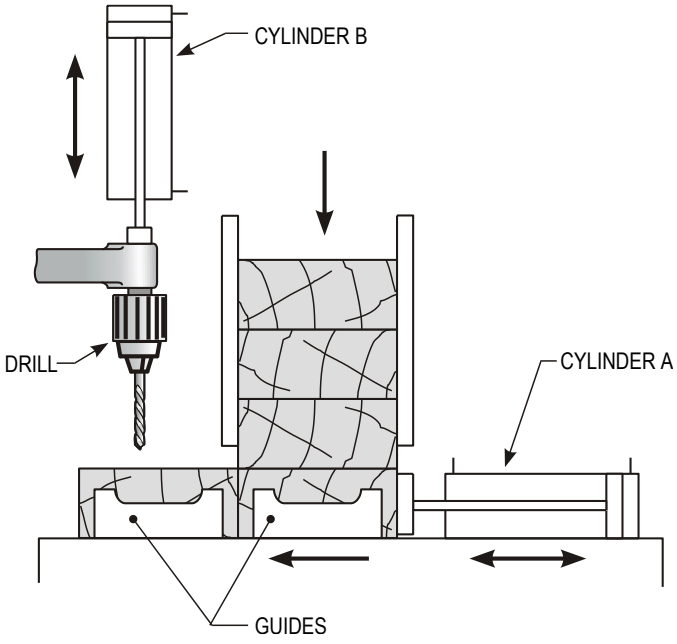
Build and test the pneumatic circuit in the diagram above. Write all answers in jotter.

a) Name the components labelled Valve D, Valve F and Valve H.
b) If Valve H was removed from the circuit, explain the effect this would have on the operation of the furnace door.
c) Using appropriate terminology, explain how the circuit operates from when Valve A is pressed.
d) A short delay is required before Cylinder B goes positive. Alter your circuit to take this into account.

## 9.

A company has automated a small section of its production line. The system involves a component being pushed into position by a cylinder, drilled and then released ready to begin again. The system involves two cylinders working together in sequence: A+, B+, A-, B-. The system is automatic and should continue to work through the sequence. However, a start/stop button should be incorporated that would allow the system to be shut down in the event of a jam or the drill bit breaking.

a) Design a pneumatic system that would carry out this sequence of operations. Write all answers in jotter.
b) Build and test your solution.
c) Using appropriate terminology, describe how it completes one sequence.
d) Explain your choice of actuators that sense the fully instroked and outstroked positions of the two pistons.

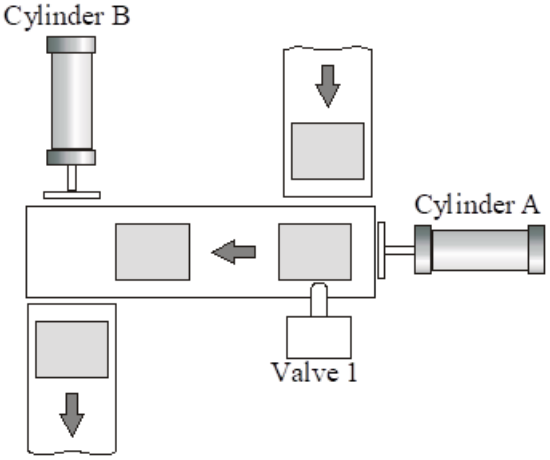


**10.** Engraving is an operation that requires great skill and is very time consuming. In an attempt to speed up the process, two double acting cylinders are needed to work together in sequence to position the component and hold it in place. Cylinder A pushes the component to be engraved into position. The clamping process is achieved by connecting a parallel linkage system to the end of cylinder B. When the cylinder instrokes, the linkage closes around the component to hold it steady.

The complete sequence required is: **A +, B –, A –, B +**

a) Design a pneumatic circuit to carry out this sequence. Write all answers in jotter.
b) Build and test the circuit and ensure that it operates correctly.
c) It is felt that the clamping operation works too quickly and sometimes the component is moved out of line. Alter your circuit so that cylinder B instrokes more slowly.
d) Not enough time is allowed for the engraving operation to take place before the component is released. A short delay is needed in the circuit. Using the correct components, position the delay where you think it would be most effective. Explain your reasons for inserting it there.

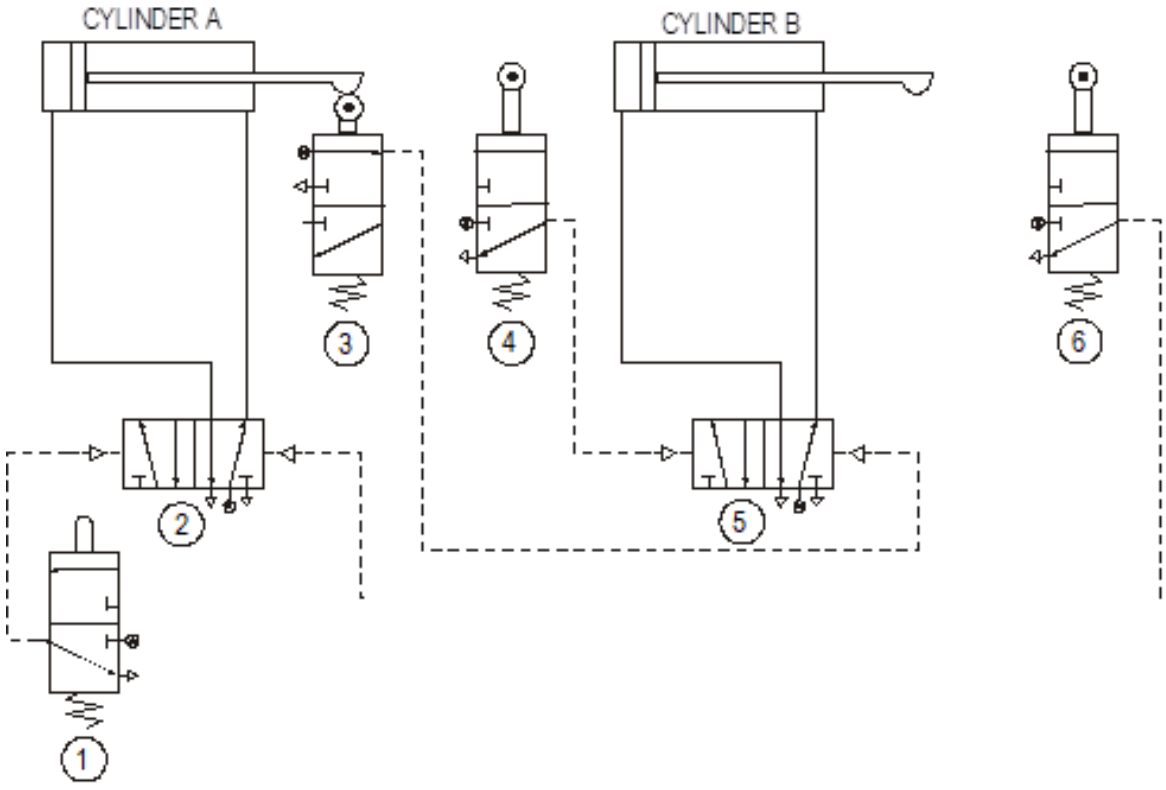
**11.** A pneumatic system is used to transfer packages between conveyor belts as shown. The pneumatic circuit is also shown.




The sequence of operation of the cylinders is:

**A +, B +, A –, B –**

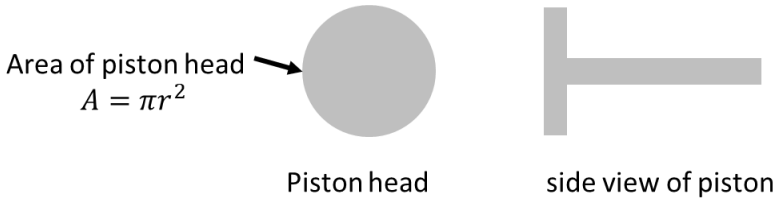
a) Build and test this circuit. Write all answers in jotter.
b) Name valves 1, 2 and 4.
c) In your jotter, describe how the circuit operates.
d) If the packages were too light to actuate valve 1, describe another way to detect the packages.



## Forces in a Single Acting Cylinder

When a single acting cylinder outstrokes, it produces a force. We can use this force to lift objects or push them around. The size of the force produced by the cylinder as it outstrokes depends on two things - the air pressure supplied to the cylinder and the surface area of the piston. This means that we can increase the force by either increasing the air pressure or increasing the size of the piston.

Air pressure is measured in bar or in N/mm². 1 bar is equal to 10 N/mm². N/mm² is the preferred unit of measure.




**Worked example**

A piston of Diameter 25mm has a pressure of 4N/mm². Calculate the force produced as the piston out strokes.

P = 4

$A = \pi r^2$ = 3.14 x 12.5 = 491mm²

F = PA

F = 4 x 491 = 1964N

= 1.96kN

## Forces in a Double Acting Cylinder

Calculating forces in a double acting cylinder involves different calculations depending on whether the piston is in stroking or out stroking. If the piston is out stroking use the same formula as for a single acting cylinder. F = PA

If the piston is in stroking, then the formula is different. As you can see in the diagram below the surface having pressure applied is smaller due to the piston rod.




Therefore, you must subtract the area of the end of the piston rod from the main area of the piston head

**Worked example**

A piston of Diameter 25mm and the rod a diameter 0f 5 mm. A pressure of 4N/mm² is applied. Calculate the force produced as the piston out strokes and in strokes. The formula for in stroking D.A.C. is F = P x (A1 – A2)

P = 4

$A1 = \pi r^2$ = 3.14 x 12.5 = 491mm²

$A2 = \pi r^2$ = 3.14 x 2.5 = 19.6mm²

Out stroking piston F = 4 x 491 = 1964N =1.96kN

In stroking piston F = 4 x (491 – 19.6) = 1885.6N = 1.89kN

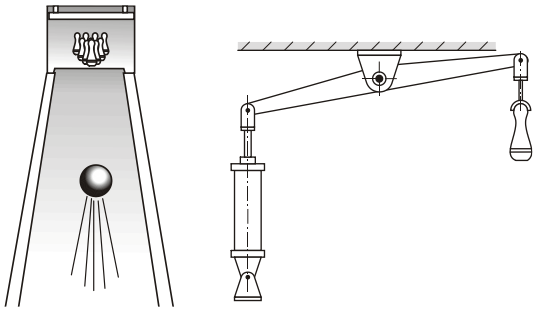
# Interfacing and Electronic control

Although pneumatic circuits have many advantages, they can become complicated and expensive when lots of components are needed. They can also be difficult to set up and control. One possible way of overcoming these problems is to use electronics or a computer interface to control the operation of pneumatic circuits. The advantage of this is that electronic signals can be transmitted over much greater distances than pneumatic signals. However, the main advantage is that electronic signals respond faster than pneumatic signals and use less energy.

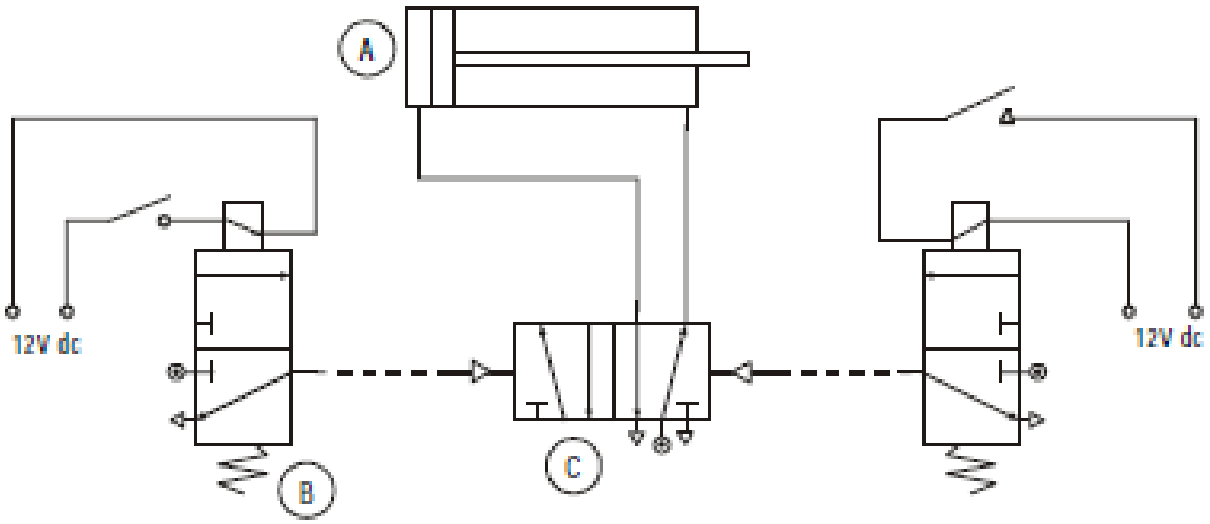
To control a pneumatic system electronically, we require the use of a solenoid operated 3/2 valve. This valve is actuated when an electric current energises the coil of the solenoid. If we can control the current flowing to the coil, then we can control the operation of the entire circuit. This can be achieved easily by connecting switches or sensors in series with the solenoid.

## Example

A ten-pin bowling complex uses double acting cylinders to set up the skittles once they have been knocked down. The cylinder is controlled by an electric switch arrangement which energises solenoid operated 3/2 valves.



The circuit diagram is shown below.




**12.** In your jotters, describe how this circuit operates.

1) Build and test the electrical and pneumatic circuit shown in diagram above.

a) What advantages are there of controlling pneumatic circuits electronically?

b) If only pneumatic components were available, redesign the circuit so that it would carry out the same operation.

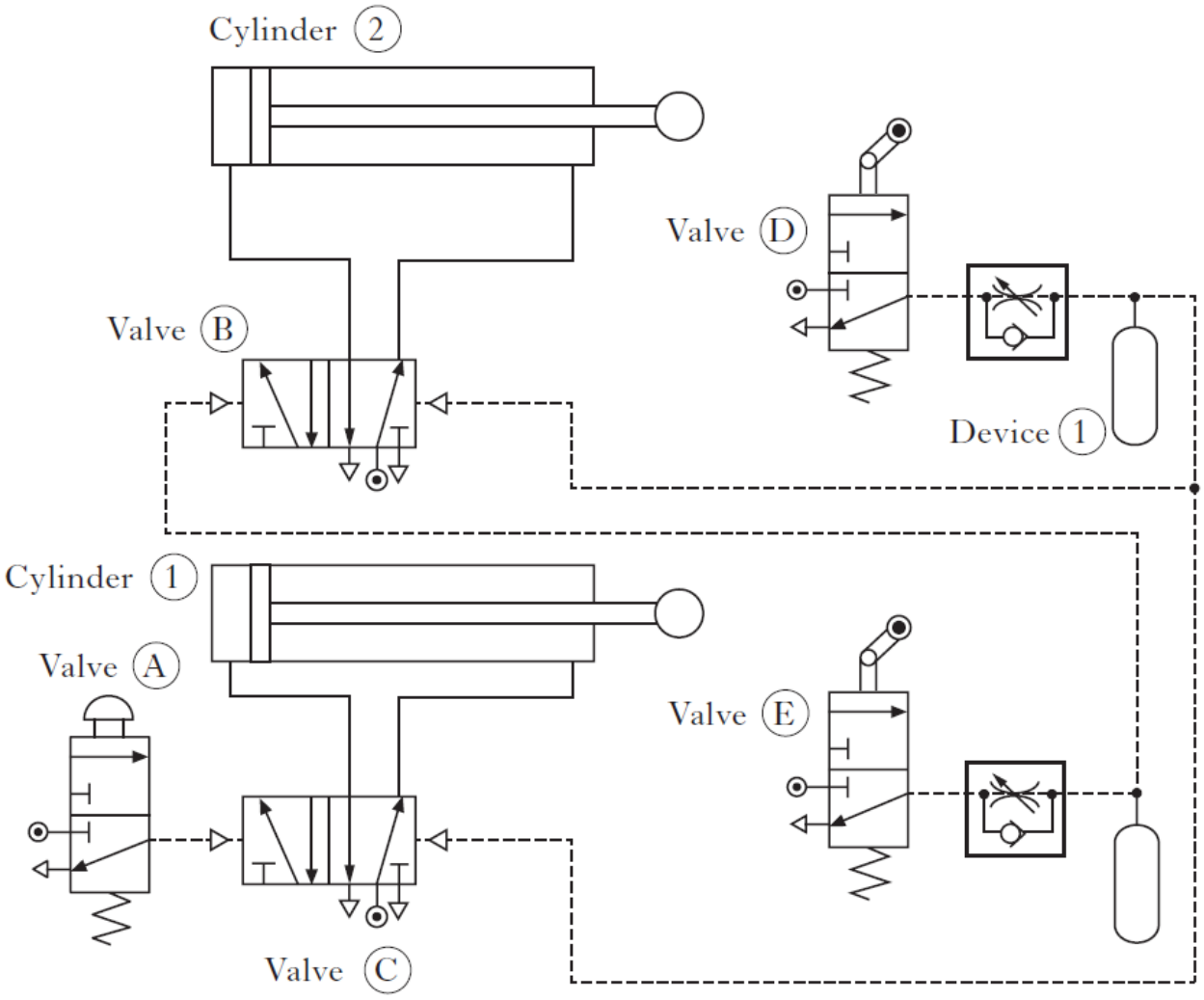
**13.** A waste disposal system makes use of a hopper and container. The container moves forward under the hopper, which tips up to empty its contents. Limit switches are used to detect if the container is in place and if the hopper has been tipped up properly. The system needs a start/stop button that will allow overall control of the circuit.

a) Design an electronic and pneumatic system to fully solve this problem.
b) Using appropriate terminology, explain how the circuit operates.
c) Give examples of other 'sensors' that could be used to detect if the container and hopper are in the correct place.
d) It is discovered that once the hopper has tipped right over, the double acting cylinder begins to instroke before all the contents have emptied out. Insert a time delay into the circuit that will overcome this.

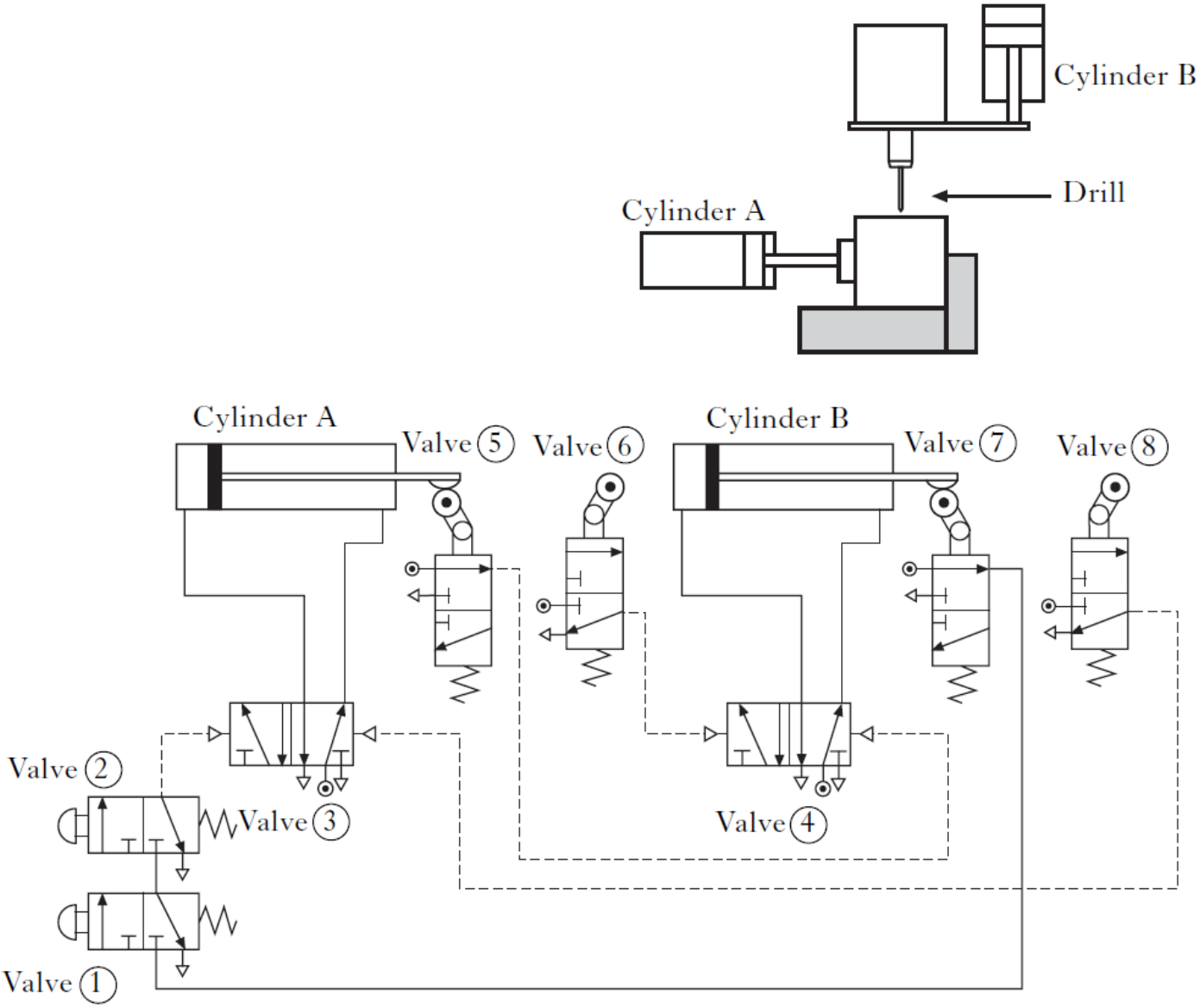
## Exam style questions



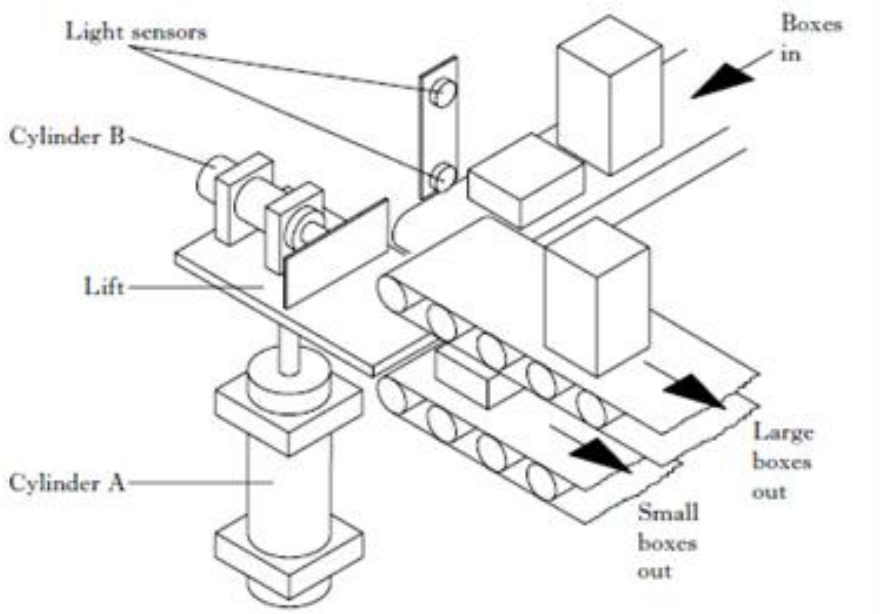
**14.** A pneumatic system controls the movement of a 'low-rider' car. When activated, two cylinders raise the front and rear of the car. Describe, using appropriate terminology, the operation of the circuit.



**15.** A pneumatic circuit is used in a drilling operation. Describe, using appropriate terminology, the operation of the circuit.



**16.** A process engineering company has been asked by an online retailer to develop a mechanical system to automatically sort boxes by size and direct them onto the appropriate conveyor for transfer to the despatch area.

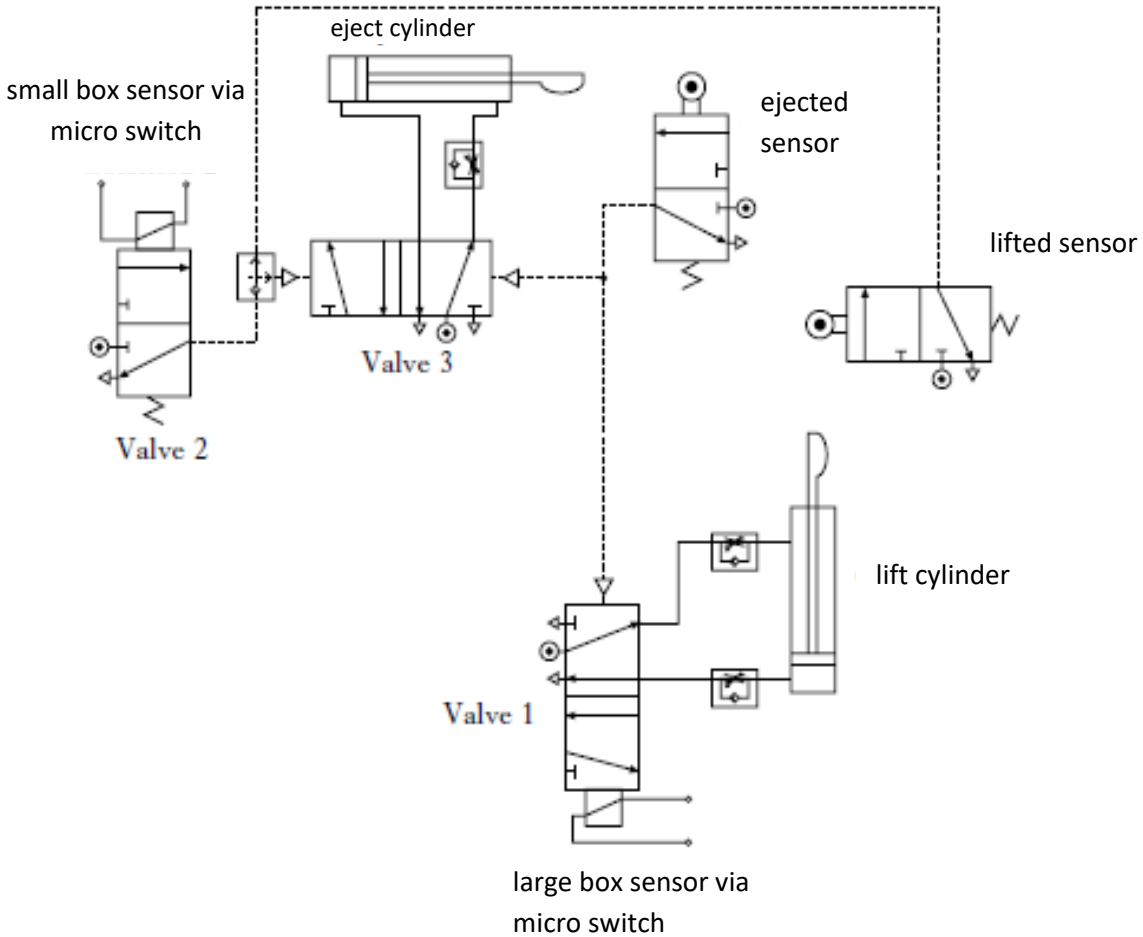



**Assignment problem brief**

Design and model a pneumatic-based system to control the cylinders in the sequence as shown in the table below.

| Box sensed | Cylinder Sequence |
|---|---|
| Small boxes | B+ B- |
| Large Boxes | A+ B+ B- A- |

**Potential answer**

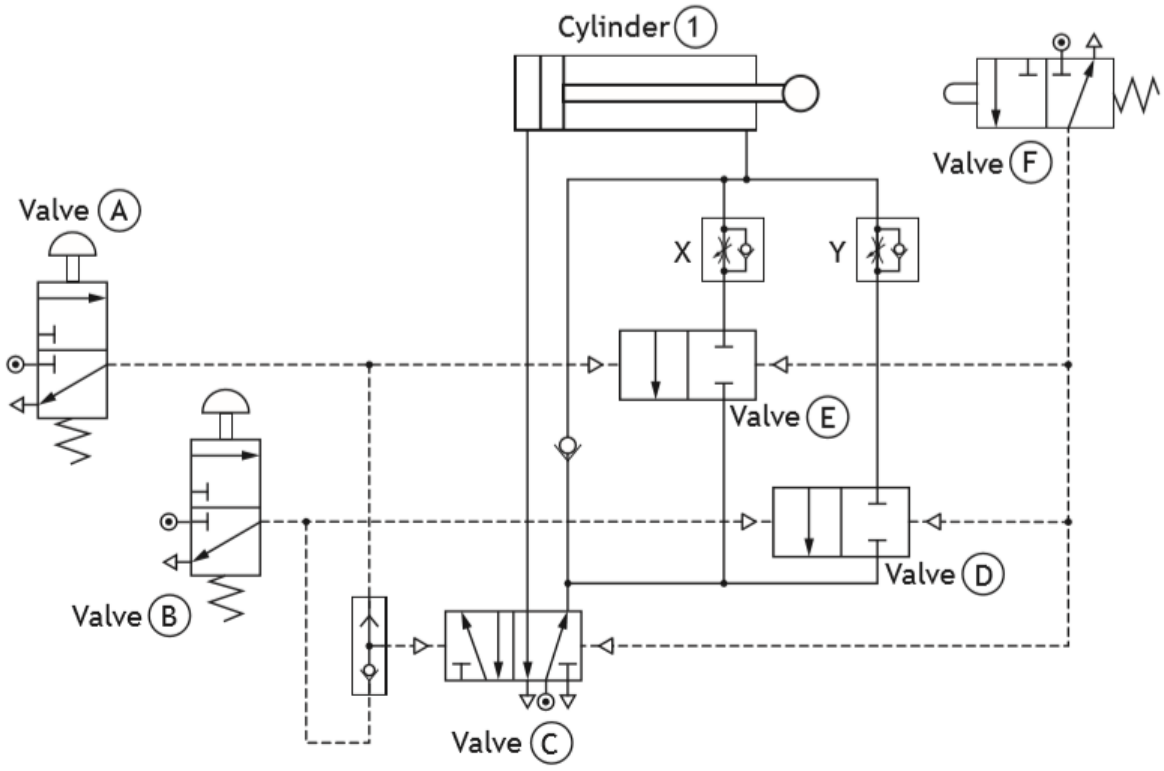



**Q9 from 201**

**9.** A pneumatic circuit is used to compress two different types of material in the production of children's car seats. An operator actuates either valve (A) or (B) when the material is in position.

Components X and Y are set at different levels.

Valves (D) and (E) are 2/2 valves. They allow air to flow through when actuated in one state but not when they are in the other state.




(a) Describe, with reference to **all** of the components in the pneumatic circuit, the operation of the system when valve (A) is pressed and released then valve (B) is pressed and released. **7**

**9. (continued)**

(b) Explain the effect that pressing **both** valves (A) and (B) together would have on the outstroking speed of the cylinder. **2**